

Opportunism and its Critics

Oliver E. Williamson

University of California, Berkeley, CA, USA

Although there is growing agreement that bounded rationality is the appropriate cognitive assumption for describing economic organization,¹ there is less agreement on how the self-interestedness of economic actors should be described. Transaction cost economics has proposed that economic agents be described as opportunistic, where this contemplates self-interest seeking with guile. That has turned out to be a controversial formulation.

I begin with a brief discussion of behavioral assumptions, including an explanation of why opportunism is centrally implicated. Recent approaches to economic organization that emphasize trust are considered in the following section. The effects of suppressing opportunism are examined in the third section and conclusions follow.

BEHAVIORAL ASSUMPTIONS

Herbert Simon concluded his 1984 address to the American Political Science Association with the following observation (1985, p. 303) :

Nothing is more fundamental in setting our research agenda and informing our research methods than our view of the nature of the human beings whose behavior we are studying. It makes a difference, a very large difference, to our research strategy whether we are studying the nearly omniscient *Homo economicus* of rational choice theory or the boundedly rational *Homo psychologicus* of cognitive psychology. It makes a difference to research, but it also makes a difference for the proper design of political institutions. James Madison was well aware of that, and in the pages of the *Federalist papers* he opted for this view of the human condition (*Federalist*, No. 55):

As there is a degree of depravity in mankind which requires a certain degree of circumspection and distrust, so there are other qualities in human nature which justify a certain portion of esteem and confidence.

—a balanced and realistic view, we may concede, of bounded human rationality and its accompanying frailties of motive and reason.

Opportunism corresponds to the frailty of motive 'which requires a certain degree of circumspection and distrust' in the transaction cost economics scheme of things.

To be sure, economizing on bounded rationality is the predominant concern for many problems of economic organization. W. Ross Ashby's interesting examination of the design for a brain (1960) is a strictly bounded rationality construction. Simon's (1962) examination of the architecture of complexity proceeds similarly. The same is true of team theory (Marschak and Radner, 1972). Ascribing benign intentions to one of the parties to a contract, even though the other is opportunistic, also simplifies the analysis of contracting (Hurwicz, 1973).

The merits of this research notwithstanding, huge numbers of interesting problems of economic organization are missed or misconstrued if opportunism is ignored or suppressed. But for opportunism, most forms of complex contracting and hierarchy vanish. That is because contractual incompleteness (due to bounded rationality) need never give rise to contractual difficulties if parties to a contract can be relied on to self-enforce all agreements. Incompleteness notwithstanding, all gaps, omissions, errors, etc. will be cured—in the sense that autonomous contracting parties will quickly and assuredly be restored to the contract curve following any unanticipated disturbance—by introducing a 'general clause'. Thus although parties to a contract are assumed to realize all of the advantages that their original positions (e.g. monopoly ownership of resources) entitle them, *no ex post maladaptation problems will arise* if (1) the parties promise at the outset to 'disclose all relevant information candidly and to behave in a co-operative way during contract execution and at contract renewal intervals' and (2) these promises are self-enforcing. Such promises are easy to make. If, however, the 'force of

words . . . [are] too weak to hold men to the performance of their covenants' (Hobbes, 1928, p. 92), then such promises are not self-enforcing.

My insistence that opportunism be accorded co-equal status with bounded rationality does not imply that I believe that most economic agents are engaged in opportunistic practices most of the time. Rather, most economic agents are engaged in business-as-usual, with little or no thought to opportunism, most of the time. That opportunism does not continuously intrude is partly because many economic agents are well-socialized. The discriminating alignment of governance structures in relation to the hazards of opportunism—whereby opportunism is mitigated—is also a contributing factor. Note with respect to both of these points that the need to protect a (well-socialized) majority against the predatory tendencies of a determined minority is an important reason to provide contractual safeguards. H. L. A. Hart's remarks are pertinent (1961, p. 193; emphasis in original):

Neither understanding of long-term interest, nor the strength of goodness of will . . . are shared by all men alike. All are tempted at times to prefer their own immediate interests . . . 'Sanctions' are . . . required not as the normal motive for obedience, but as a *guarantee* that those who voluntarily obey shall not be sacrificed by those who would not.

EMBEDDEDNESS AND TRUST²

Many sociologists take exception with opportunism. Some refer to the transaction cost economizing program that is associated with opportunism as neo-Hobbesian (Granovetter, 1985, pp. 494–5). A different research agenda is implied if 'social relations, rather than institutional arrangements [governance structures] or generalized morality, are mainly responsible for the production of trust in economic life' (Granovetter, 1985, p. 491). According to the social relations approach, an understanding of economic organization resides in studying 'embeddedness' rather than the governance of contractual relations.

Granovetter evidently holds that transaction cost economics is unable or unwilling to come to grips with embeddedness. As it turns out, however, transaction cost economics is able to accommodate at least some of the more important embeddedness

conditions to which Granovetter refers by working out of the two-level approach to institutional economics described by Lance Davis and Douglass North (1971, pp. 6–7). The distinction here is between the 'institutional environment' and the 'institutions of governance', where the former is concerned with politics, law, the judiciary, norms, customs and the like, and the latter examines the institutions of private ordering—markets, hybrids, hierarchies, bureaucracies and the like. The institutional environment describes the macrostructure; the institutions of governance deals with the microstructure. Changes in the institutional environment change the absolute and, often, the comparative costs of governance—often in predictable ways (Williamson, 1991a). Rather, therefore, than ignore or dispute the embeddedness conditions to which Mark Granovetter (1985), Ronald Dore (1985), Lynne Zucker (1986) and other sociologists refer, transaction cost economics treats the institutional environment and the institutions of governance in a combined way. Societal devices for attenuating opportunism are therefore valued; but that is true of *all* devices that economize on transaction costs. Codes or other communication aids that economize on bounded rationality, for example, are put on the very same footing.

Opportunism engages the *realpolitik*s of economic organization and is an unflattering behavioral assumption. Trust is sometimes treated as an antonym for opportunism and supports a more favorable opinion of human nature. Many regard it as a more instructive construction. Partha Dasgupta's treatment of 'Trust as a commodity' begins with the claim that 'Trust is central to all transactions and yet economists rarely discuss the notion' (1988, p. 49).

Trust has long been featured by sociologists, many of whom have begun to treat trust in calculative terms (Bradach and Eccles, 1989; Dore, 1983; Coleman, 1990). James Coleman is explicit on this: 'Situations involving trust constitute a subclass of those involving risk. They are situations in which the risk one takes depends on the performance of another actor' (Coleman, 1990, p. 91). According to this formulation, trust is warranted when the expected gain from placing oneself at risk to another is positive, but not otherwise. Indeed, the decision to accept such a risk is taken to imply trust (Coleman, 1990, p. 105).

This theme is repeated throughout the influential seminar series organized by Diego Gambetta and

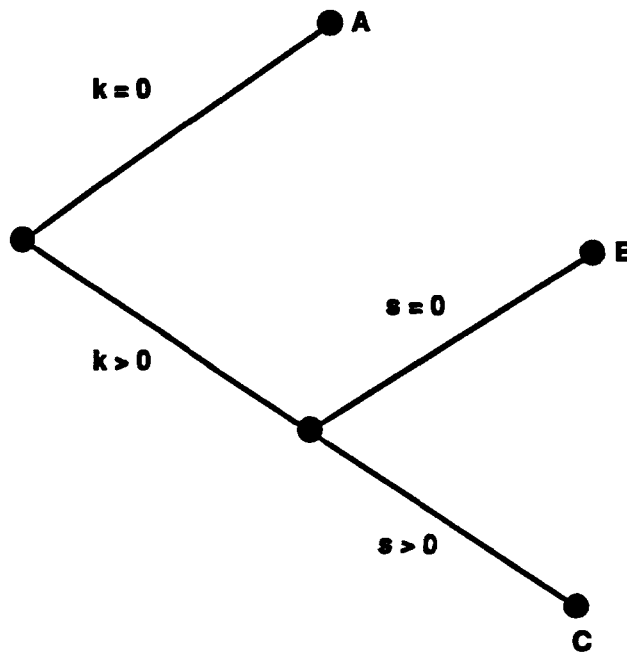
published under the title *Trust: Making and Breaking Cooperative Relations* (1988). That volume closes with the following unifying observation (Gambetta, 1988, p. 217):

... there is a degree of convergence in the definition of trust which can be summarized as follows: trust ... is a particular level of the subjective probability with which an agent assesses that another agent or group of agents will perform a particular action When we say we trust someone or that someone is trustworthy, we implicitly mean that the probability that he will perform an action that is beneficial or at least not detrimental to us is high enough for us to consider engaging in some form of cooperation with him.

Jeffrey Bradach and Robert Eccles expressly embrace this view in their recent treatment of 'Price, authority, and trust' in the *Annual Review of Sociology* (1989, p. 104).

Presumably, however, *the object is not to describe human actors in a user-friendly way but to understand complex economic organization*. I contend that calculated trust is a contradiction in terms and that the study of economic organization is better served by treating commercial transactions without reference to trust.

The simple contractual schema out of which transaction cost economics works is pertinent. Exchange is therein described as a triple (p, k, s) , where p refers to the price at which the trade takes place, k to the investment hazards that are associated with the exchange—which hazards would vanish were it



	p	k	s
Node A	p_1	0	0
Node B	$\bar{p}$	$\bar{k}$	0
Node C	$\wedge p$	$\bar{k}$	$\wedge s$

Figure 1. Simple contractual schema.

not for opportunism—and s denotes the safeguards within which the exchange is embedded. The argument is that price, hazards and safeguards are determined simultaneously.

The schematic and the values that each element in the triple takes on are shown in Fig. 1. As shown, Node A involves no hazards. The good or service in question is completely generic. Goods or services are exchanged now for prices paid now. This is the classical market exchange that Ian Macneil has described as 'sharp in by clear agreement; sharp out by clear performance' (1974, p. 738).

It will facilitate comparisons to assume that suppliers are competitively organized and are risk neutral. Prices therefore reflect an expected break-even condition. The break-even price that is associated with Node A is p_1 . There being no hazards, $k=0$. And since safeguards are unneeded, $s=0$.³

Node B is more interesting. The contractual hazard here is $\bar{k}$. If the buyer is unable or unwilling to provide a safeguard, then $s=0$. The corresponding break-even price is $\bar{p}$.

Node C poses the same contractual hazard, namely $\bar{k}$. In this case, however, a safeguard in amount $\hat{s}$ is provided. The break-even price that is projected under these conditions is $\hat{p}$. It is elementary that $\hat{p} < \bar{p}$.⁴

In the language of risk, Node A poses no risk, whereas Node B poses a risk and for Node C the risk has been mitigated. In the language of trust, trust is unneeded for Node A, whereas Nodes B and C are the low trust and high trust outcomes, respectively.

Bradach and Eccles contend that 'mutual dependence [i.e. $k > 0$] between exchange partners . . . [promotes] trust, [which] contrasts sharply with the argument central to transaction cost economics that . . . dependence . . . fosters opportunistic behavior' (1989, p. 111). What transaction cost economics says, however, is that because opportunistic agents will not self-enforce open-ended promises to behave responsibly, efficient exchange will be realized only if dependencies are *supported* by credible commitments. Wherein is trust implicated if parties to an exchange are far-sighted and reflect the relevant hazards in the terms of the exchange? (A better price ($\hat{p} < \bar{p}$) will be offered if the hazards ($k > 0$) are mitigated by cost-effective contractual safeguards ($\hat{s} > 0$.) More generally, I maintain that credible commitment is a more exact and accurate way to describe commercial exchange. Reference to trust in this connec-

tion is redundant at best and, because it is diffuse and fuzzy, often promotes confusion.

Further, while credible commitments deter breach and support more efficient exchange, breach is not wholly precluded. On the contrary, if it is inefficient to supply under some state realizations, then an optimal contract will project breach for those states. Whereas *efficient breach* of commercial contract is easy to reconcile with a calculative approach to contract, the notion that trust can be efficiently breached experiences considerable strain. Much of the contract law literature would be clarified if trust were consistently used in a delimited way.

SUPPRESSING OPPORTUNISM

Economic organization is unarguably complex, and simplification is vital. Suppressing opportunism can be thought of as a simplifying move. Provided that analysts who suppress opportunism do this knowingly and come back to assess the ramifications, who could object?

I submit, however, that opportunism is more often suppressed unknowingly or selectively and that, once done, the ramifications are rarely assessed. That opportunism was so long suppressed unknowingly is because that is how most students of economics learned their subject. As of the early 1970s, the standard economic models treated 'individuals' as playing a game with fixed rules which they obey. They do not buy more than they can pay for, they do not embezzle funds, they do not rob banks' (Diamond, 1971, p. 31).

To be sure, that changed as conditions of adverse selection, moral hazard and shirking were addressed by information economics (Rothschild and Stiglitz, 1976) and agency theory (Alchian and Demsetz, 1972; Holmstrom, 1979). Moreover, the resulting treatments of separating equilibria and efficient *ex ante* incentives for risk-averse agents were (and are) instructive. But there are three concerns:

- (1) Adverse selection and moral hazard are technical terms that needlessly truncate the study of economic organization as compared with the range of phenomena that are implicated by opportunism;
- (2) Relatedly, managerial discretion is a more general phenomenon than shirking; and

- (3) Uncritical assessment of naive (especially utopian) forms of organization is encouraged by the failure to make prominent and insistent provision for opportunism.

Truncation

Opportunism is a less technical term than adverse selection and moral hazard. It suggests, correctly, that the troublesome behavior in question is not an arcane economic condition but is familiar and pervasive. Not only are the failures to self-disclose true attributes *ex ante* (adverse selection) and true performance *ex post* (moral hazard) both subsumed under opportunism, but the failure to tell the truth, the whole truth and nothing but the truth is implicated by opportunism. The possibilities that economic agents will lie, cheat and steal are admitted. The possibility that an economic agent will conform to the letter but violate the spirit of an agreement is admitted. The possibilities that economic agents will deliberately induce breach of contract and will engage in other forms of strategic behavior are admitted. More generally, the unapologetic reference to opportunism invites attention to and helps to unpack a much wider set of phenomena than normally arise when reference is made to adverse selection and moral hazard.

Managerial Discretion

The standard agency theory setup entails incentive alignment between a principal and an agent in a situation characterized by differential information (where the agent normally enjoys the advantage) and differential risk aversion (where the agent is ordinarily assumed to be more risk averse). Thus let the output of the agent be $X = X(e, \theta)$, where e refers to the effort by the agent and θ is the state realization. The principal, who is assumed to be risk neutral, is able to observe output only, while the agent knows the state realization and chooses how much effort to expend. The principal's problem is to devise a cost-effective sharing rule in which provision is simultaneously made for the incentive of the agent to be productive (which favors a high sharing rate) and the premium that must be paid to the agent to compensate for added risk (which favors a low sharing rate). As previously indicated, all of the relevant contracting action is concentrated in the *ex ante* incentive alignment stage of contracting.

Although agency theory has been instructive for examining a variety of labor-contracting issues (of which the compensation of a travelling sales force is an example) has been very influential in accounting (Baiman, 1982), and has been progressively generalized (Holmstrom and Milgrom, 1991), it is nonetheless a very special model upon which to base a theory of complex organization (Rosen, 1988; Hart, 1990, 1991). For one thing, to ascribe differential risk aversion to buying and supplying firms that are contracting with one another in the intermediate product market is often contrived. More importantly, the assumption of comprehensive contracting, according to which there are no *ex post* surprises, hence no attendant needs to adapt to surprises, is implausible. Relatedly, the assumption that contracts are costlessly enforceable in unrealistic. Even the public observability of outcomes can be questioned (Sappington, 1991, p. 49).

The preoccupation with effort effects (shirking) to the neglect of other forms of managerial discretion—in which power, status, job security and the like are featured—is also highly restrictive. As Oliver Hart remarks (1991, pp. 13–14, emphasis in original):

A traditional incentive scheme works well if the only agency problem comes from the fact that management dislikes working hard. However, it is much less effective if the manager obtains large control rents from running the firm (control rents which he cannot be charged for adequately *ex ante*, either because he has limited wealth or because he is risk-averse and the rents are uncertain); these control rents may represent the utility he gets from his job or from presiding over a large and perhaps growing empire, or they may represent the monetary and nonmonetary perks which the manager can obtain by virtue of his position of power. The reason incentive schemes may be less effective under these conditions is that a *very* large incentive payment may be required to induce management to give up these control rents. It may be cheaper for investors to resort to alternative mechanisms which *force* management to yield control or to curb its empire-building tendencies.

The move from incentive instruments to forcing instruments is therefore proposed (Hart, 1991). That introduces *ex post* governance considerations.

The upshot is that a useful but narrow conception of the problem of subgoal pursuit (shirking)

gives way to a more general concern (managerial discretion). Not only is attention shifted from workers to include managers, but a wider class of purposive activity comes under scrutiny and a wider set of instruments is implicated. Although reference to opportunism is not entirely responsible for those results, it has been a contributing factor.

Socialism

Whereas it was once customary to focus on 'market failures' and to discuss these in technical terms, that is untenable if the core source of failure is the human condition rather than technology. Once that is granted, then *all* forms of organization are subject to failure and the only way to proceed is comparatively (Coase, 1964). That took a very long time to register. Indeed, some continue to believe or aver that there are benign forms of organization.

The socialist controversy The socialist controversy pitted Oskar Lange and Abba Lerner against Friedrich Hayek and Ludwig von Mises. Who won the socialist controversy and why?

Although Schumpeter took exception to the application of neoclassical economic theory to capitalism, he did not question its adequacy for assessing socialism. His response to the query 'Can socialism work?' was 'Of course it can' (1942, p. 167). There was, in his judgment, 'nothing wrong with the pure theory of socialism' (Schumpeter, 1942, p. 172). Abram Bergson concurred: 'There can hardly be any room for debate; of course socialism can work. On this, Lange certainly was convincing' (1948, p. 447). As Joseph Persky remarks, 'the general consensus held that Lange had won the debates with his formal arguments from welfare economics' (1991, p. 230, n. 1).

But is allocative efficiency, which is a technical criterion, the main problem with socialism or is opportunism/managerial discretion/bureaucratization more central? To his credit, Lange raised (but thereafter dismissed) the crucial issue of bureaucratization (Lange, 1938, pp. 109–10; emphasis in original):

There is also the argument which might be raised against socialism with regard to the efficiency of public officials as compared with private entrepreneurs as managers of production. Strictly speaking, these public officials must be compared

with corporation officials under capitalism, and not with private small-scale entrepreneurs. The argument thus loses much of its force. The discussion of this argument belongs to the field of sociology rather than of economic theory and must therefore be dispensed with here. By doing so we do not mean, however, to deny its great importance. It seems to us, indeed, that *the real danger of socialism is that of a bureaucratization of economic life, and not the impossibility of coping with the problem of allocation of resources*. Unfortunately, we do not see how the same, or even greater, danger can be averted under monopolistic capitalism. Officials subject to democratic control seem preferable to private corporation executives who practically *are responsible to nobody*.

This formulation is important in three respects. First, and most important, Lange expressly entertains the possibility that the efficacy of socialism turns less on realizing the requisite technical conditions for efficient resource allocation than it does on bureaucratization. Second, Lange invokes a comparative institutional test: is bureaucratization really worse under socialism than it is under monopoly capitalism? Third, he observes that the study of bureaucracy is outside the scope of economics and belongs to sociology.

Assume, *arguendo*, that Lange is correct in identifying bureaucratization as the real danger to socialism. Is he also correct in his comparison of capitalism and socialism in bureaucratization respects?

Monopoly capitalism was perceived to be a much more serious problem in the 1930s than it has been subsequently. The 1930s were not only the period of the Great Depression (replete with stock market crashes, unemployment, the National Recovery Act, the Temporary National Economic Committee, and the like), but it was also the period when the theory of monopolistic competition (Chamberlin, 1932; Robinson, 1933) flourished. The public policy importance of monopolistic competition turned out to be limited, however—partly because deviations from hypothetical efficiency are neither here nor there, as a public policy matter, unless a superior operational alternative can be described, and partly because Schumpeterian 'handing on' works in the long run—where the process of handing on always works 'through a fall in the price of the product to the new level of costs' (Schumpeter, 1947, p. 155) whenever rivals are alert to new opportunities and

are not prevented by purposive restrictions from adopting them.

Whether capitalism, or even monopoly capitalism, is more or less troublesome than socialism in bureaucratic respects turns importantly on incentive intensity differences. Comparisons between capitalism and socialism in terms of differential appropriability and the efficacy of competition in the capital market are pertinent. The very real limitations of large capitalist, hierarchical firms notwithstanding, the socialist program is even more limited in both respects (Williamson, 1991b).

If, moreover, sociologists do not have answers to the problem of bureaucracy, it is no solution to relegate the bureaucracy problem to them. An assessment of bureaucracy needs to begin, I submit, with a statement of the behavioral assumptions. Bounded rationality manifests itself under both socialism and capitalism in terms of incompleteness—incomplete contracting under capitalism; incomplete planning under socialism. In as much as planning is the more ambitious of the two, socialism is arguably at a disadvantage in this respect.

Opportunism also manifests itself under both socialism and capitalism. Again, the behavior in question is better described as managerial discretion than shirking. Given the added instruments—especially those that relate to finance, including competition in the capital market—to which capitalism has access, the bureaucratic hazards of socialism are arguably the more severe (see below). Lange and others simply ignored the control instruments: 'Control plays no role in the socialist controversy' (Ward, 1967, p. 37). That was an egregious lapse—as subsequent events have shown.

Feasible financial reform Efforts to rehabilitate socialism by introducing the market-like features of capital markets have begun to appear. D. M. Nuti's 1989 article on 'Feasible financial innovation under market socialism' is a recent example.⁵

Interestingly, the efficacy of Nuti's plan for revaluing assets to reflect true economic values and realizing added productivity assumes a 'successfully reformed' socialist economy, the entire discussion of which is concentrated in two sentences of text (Nuti, 1989, pp. 94–5; emphasis added) :

[E]nterprises are engaged in production and trade through contractual relations with other state agencies, while planning is confined to macroeconomic policies and *truly parametric*

(that is, non-enterprise specific) instruments for the *central manipulation* of market signals. Sectoral policies can be undertaken by the government, but sector-specific subsidy on tax differentials must be applied by the government *consistently and predictably*.

Two things are worth noting about the successfully reformed socialist economy described by Nuti: (1) the description is very brief, and (2) Nuti is evidently very sanguine as to its efficacy. Lacking institutional supports, the prescription appears to assume the abolition of opportunism by agencies of the state. That simplifies the organization design problem enormously.

The key features on which Nuti relies are parametric instruments that are manipulated by a central authority in a consistent and predictable way. That is tantamount to *credible selective intervention*. Unless, however, the absence of opportunism can be credibly ascribed to central authorities, that is implausible: the same impossibility of selective intervention that applies within firms (Williamson, 1985, Chapter 6) applies likewise to governments.

Workers' management⁶ Problems of economic reform in Eastern Europe have brought renewed attention to worker-management modes of economic organization (Weisskopf, 1991). The concern is that reform economies will uncritically adopt capitalist modes of economic organization in which private ownership is emphasized. Rather than permit finance to have a significant say in the oversight of and strategic decisions of the firm, the devolution of control over state enterprises 'to communities of citizens and/or workers rather than conventional privatization' is proposed as a viable and superior alternative (Weisskopf, 1991, pp. 5, 23–67).

The *sine qua non* for the worker-managed firm is that control by equity ownership be prohibited. The definition of a worker-managed firm by John Bonin and Louis Putterman (1987) goes precisely to this issue. They define a worker-managed enterprise as:

... a productive enterprise the ultimate decision-making rights over which are held by member-workers, on the basis of equality of those rights regardless of job, skill grade, or capital contribution. A full definition would state that no non-workers have a direct say in enterprise decisions, and that no workers are denied an equal say in those decisions. This definition does not imply that any particular set of decisions must be made

by the full working group, nor does it imply a particular choice rule, such as majority voting. It says nothing about financing structures other than that financiers are not accorded direct decision-making powers in the enterprise by virtue of their non-labor contributions, and it does not say anything about how income is distributed among workers. On all of these matters, all that is implied is that ultimate decision-making rights are vested in the workers, and only in the workers. Thus, the basic definition centers on an allocation of governance rights, and is simultaneously economic and political (p. 2).

This definition does not preclude hierarchical structure, specialized decision making, a leadership elite or marginal-product payment schemes. It merely stipulates that finance can have no decision rights in the labor-managed enterprise. The question is whether these financial restrictions come at a cost.

Bonin and Putterman evidently believe that they do not. That, however, is because they operate out of a technological (firm as production function) rather than contractual (firm as governance structure) setup. The former focuses on the marginal conditions for efficient resource allocation (McCain, 1977) and suppresses the differential contractual hazards that are associated with alternative forms of governance. Indeed, governance issues never arise, hence cannot be assessed, within this orthodox framework. If, however, a critical—indeed, I would say, *the* critical—attribute of equity is the ability to exercise contingent control by concentrating votes and taking over the board of directors, then McCain's demonstration that allocative efficiency is identical under standard equity and risk participation bonds is simply inapposite.⁷

Indeed, if risk-participation finance is available on more adverse terms than standard equity because holders are provided with less security against mismanagement and expropriation, then the constraints that Bonin and Putterman have built into the worker-managed firm come at a cost.⁸ To be sure, the worker-managed firm may be able to offset financial disabilities by offering compensating advantages. If those advantages are not uniform but vary among firms and industries, then the net gains of the worker-managed firm will vary accordingly.

Firms that can be mainly financed with debt are the obvious candidates for worker-management. That is because if there is little equity-like capital at stake, then there is little reason for equity to ask or expect that pre-emptive control over the board of

directors will be awarded to equity as a contractual safeguard. The question then is what types of firms best qualify for a preponderance of debt financing?

The partnership form of organization works well in professional organizations, such as law and accounting firms, where the need for firm-specific physical capital is small. There being little need for equity capital to support investment in such firms, the control of these firms naturally accrues to those who supply specialized human assets (Williamson, 1989, pp. 24–6; Hansmann, 1990). Also, peer group forms of organization can and do operate well in *small* enterprises where the membership has been carefully *screened* and is committed to democratic ideals (Williamson, 1975, Chapter 3). These exceptions aside, 'third forms' experience serious incentive disabilities.⁹

The evidence from Eastern Europe is pertinent. Maciej Iwanek (1991) remarks of the Polish experience that 'except [among] advocates of workers' management, nobody believes that the . . . governance scheme of state-owned enterprises [by workers' management] creates strong incentives' (1, p. 12); Manuel Hinds (1990) concludes that 'absenteeism, shirking, and lack of initiative are pervasive in the self-managed firm' (p. 28); and Janos Kornai (1990) counsels that 'it would be intellectually dishonest to hide the evidence concerning the weakness of third forms' (p. 144).

With respect to the above, (1) the socialist controversy, (2) the benign use of financial instruments and (3) the purported efficacy of workers' management are all unified by one thing: a disregard for the organizational and performance ramifications of opportunism. The suppression of opportunism, however, is an exercise in King Canute economics. Had that naivete been recognized and confronted, the 'experiment' to which Boris Yeltsin has recently referred might have been run differently.¹⁰

I think that [the] experiment which was conducted on our soil was a tragedy for our people and it was too bad that it happened on our territory. It would have been better if the experiment had been conducted in some small country, at least, so as to make it clear that it was a utopian idea, although a beautiful one.

CONCLUSIONS

Suppose, *arguendo*, that opportunism is conceded to be the appropriate way to describe self-interest seeking. What are the consequences?

The myopic response, which is associated with Machiavelli, is to engage in pre-emptive opportunism. Out of awareness that other economic actors will break promises when it suits their purposes, the prince is advised to do the same (Gauss, 1952, pp. 92–3).

Reciprocal or pre-emptive opportunism is not the only lesson to be gleaned from an awareness that human agents are not fully trustworthy. Indeed, that is a very primitive response. Transaction cost economics emphasizes two other messages instead. First, do not contract in a naive way. Second, attempt to mitigate opportunism in cost-effective ways. These two are closely related.

Taking the hazards of contracting as given, the message is this: do not allow user-friendly terms like promise and trust to mask the objective features of the deal. Identical contracting language and oral representations notwithstanding, if one contract objectively poses greater hazards than another, then the differential hazards should be reflected in the respective price of each deal—which could include a refusal to deal if the hazards are perceived to be prohibitive.

Hazards, however, need not be taken as given. Transactions that are subject to *ex post* opportunism will benefit if cost-effective safeguards can be devised *ex ante*. Rather than reply to opportunism in kind, the wise prince is one who seeks to both give and receive credible commitments. That is a much deeper and more important contractual response, but it requires that the hazards of opportunism be faced candidly rather than suppressed.

More generally, the argument is this: superior outcomes will be realized by addressing economic organization in more veridical terms. That will be facilitated by asking and answering the question ‘What’s going on here?’ as opposed to asserting ‘This is the law here’. Although the preoccupation of social scientists with the latter is understandable, that is often accomplished by suppressing bounded rationality and/or opportunism—which invites contractual naivete and remediable error. The analytical need is to deal symmetrically with organizational failures in all of their forms—market and nonmarket alike. The unapologetic application of bounded rationality and opportunism to all forms of organization helps to expose core issues, to which economizing analysis can then be applied.

rationality implied nonrationality or irrationality. That has changed as the idea that bounded rationality implies incomplete contracting has taken hold.

2. This section is based on Williamson (1992).
3. Another way of putting it is that (transition problems aside), each party can go its own way without cost to the other. Competition provides a safeguard.
4. For a more systematic development, see Williamson (1983). For related empirical work, see Scott Masten and Keith Crocker (1985) and the surveys reported by Paul Joskow (1988, 1991).
5. The following three paragraphs follow Williamson (1991b, p. 179).
6. This subsection is based on Williamson (1989, pp. 22–4).
7. McCain and Putterman evidently believe that the reason that all projects are not debt financed is that ‘raising adequate capital for risky ventures requires an instrument that allows the investor to share not only in the risk of default, but also in the full range of potential *high* returns’ (Putterman, 1984, p. 184; emphasis in original). The risk-participation bond purportedly mimics equity because it has this upside participation feature. But there is more to equity than earnings participation. McCain and Putterman ignore the distinctive control features of equity. Putterman in 1984 seems to ignore his earlier observation that ‘if equity-owners value their voting control over firm policies, and if worker-run firms cannot (on principle) share such control with their equity owners, then the costs of raising equity will be higher for the worker-run firm’ (1982, p. 158).
8. Putterman has since discussed the worker-managed firm without reference to McCain and argues that efficiency will obtain if all capital goods are rentable (1990, p. 187). That, however, is unrealistic for durable, specialized, nonmobile assets (Williamson, 1988). Although Putterman subsequently concedes that full rental is not realistic, he remains sanguine that the worker-managed firm can combine ‘bank and debt financing with internal financing and/or equity’ in an efficient way. In that event, the ‘horizon problem’ is all that remains, and he believes that to be manageable (Putterman, 1990, p. 188).
9. The declining industry represents a separate and rather special case. If plant closings are objectively in prospect, then the nature of the bargaining relation between workers and firm can be presumed to change. Thus whereas incumbent workers benefit from prospective reputation effect features so long as continuity of the business can be projected (Williamson, 1985, pp. 259–61), this protection weakens when termination is in prospect. Outside investors may undervalue firm-specific human capital in deciding to terminate. Or they may demand give-backs as a condition of continuity. One possible way by which to orchestrate the waning years of an enterprise is for the workers to buy the investors out, on nominal terms, and arrange their own give-backs (that is, cut their own pay). (Recent sales of steel mills in the USA to workers appear to qualify. These mills were scheduled for shutdown and workers, upon taking control, cut their wages and benefits.)
10. The Yeltsin quote appears in an Associated Press

NOTES

1. It was not so long ago that bounded rationality was regarded with disdain, the belief being that bounded

interview that was published in the *International Herald Tribune*, 7–8 September 1991, p. 4.

REFERENCES

- W. R. Ashby (1960). *Design for a Brain*, New York: John Wiley.
- S. Baiman (1982). Agency research in managerial accounting: a survey. *Journal of Accounting Literature* 1, 154–213.
- A. Bergson (1948). Socialist economies. In *Survey of Contemporary Economies* (edited by H. Ellis), Philadelphia: Blackiston, pp. 430–58.
- J. Bonin and L. Putterman (1989). *The Economics of Cooperation and the Labor Managed Economy*, New York: Cambridge University Press.
- J. Bradach and R. Eccles (1989). Price, Authority, and Trust. *American Review of Sociology* 15, 97–118.
- E. H. Chamberlin (1933). *The Theory of Monopolistic Competition*, Cambridge, MA: Harvard University Press.
- R. H. Coase (1964). The regulated industries: discussion. *American Economic Review* 54, May 194–7.
- J. Coleman (1990). *The Foundations of Social Theory*, Cambridge, MA: Harvard University Press.
- P. Dasgupta (1988). Trust as a commodity. In *Trust: Making and Breaking Cooperative Relations*, (edited by D. Gambetta), Oxford: Basil Blackwell, pp. 49–72.
- L. E. Davis and D. C. North (1971). *Institutional Change and American Economic Growth*, Cambridge: Cambridge University Press.
- P. Diamond (1971). Political and economic evaluation of social effects and externalities: comment, In *Frontiers of Quantitative Economics* (edited by M. Intrilligator), Amsterdam: North-Holland, pp. 30–32.
- R. Dore (1983). Goodwill and the spirit of market capitalism. *British Journal of Sociology* 34, December, 459–82.
- D. Gambetta (1988). Can we trust trust? In *Trust: Making and Breaking Cooperative Relations* (edited by D. Gambetta), Oxford: Basil Blackwell, pp. 213–37.
- C. Gauss (1952). Introduction. *Machiavelli* (1952), pp. 7–32 (Niccolo Machiavelli, *The Prince*, New York: New American Library).
- M. Granovetter (1985). Economic action and social structure: the problem of embeddedness. *American Journal of Sociology* 91, November, 481–501.
- H. Hansmann (1988). The ownership of the firm. *Journal of Law, Economics, and Organization* 4, (Fall), 267–303.
- H. L. A. Hart (1961). *The concept of Law*, Oxford: Oxford University Press.
- O. Hart (1990). An economist's perspective on the theory of the firm. In *Organization Theory* (edited by O. Williamson), New York: Oxford University Press, pp. 154–71.
- O. Hart (1991). *Theories of Optimal Capital Structure*, forthcoming, Washington, DC: Brookings Institution.
- M. Hinds (1990). *Issues in the Introduction of Market Forces in Eastern European Socialist Economies*, The World Bank. Report No. IDP-0057.
- T. Hobbes (1928). *Leviathan, or the Matter, Forme, and Power of Commonwealth Ecclesiastical and Civil* (1651), Oxford: Basil Blackwell.
- B. Holmstrom (1979). Moral hazard and observability. *Bell Journal of Economics* 10, Spring, 74–91.
- B. Holmstrom and P. Milgrom (1991). Multi-task principal-agent analysis. *Journal of Law, Economics, and Organization* 7 (Special Issue), 24–52.
- L. Hurwicz (1973). The design of mechanisms for resource allocation. *American Economic Review* 63, May, 1–30.
- M. Iwanek (1991). Issues of institutions transformations and ownership changes in Poland. *Journal of Theoretical and Institutional Economics* forthcoming.
- P. Joskow (1988). Asset specificity and the structure of vertical relationships: empirical evidence. *Journal of Law, Economics, and Organization* 4, Spring, 95–117.
- P. Joskow (1991). The role of transaction cost economics in antitrust and public utility regulatory policies. *Journal of Law, Economics, and Organization* 7 (Special Issue), 53–83.
- J. Kornai (1990). The affinity between ownership forms and coordination mechanisms: the common experience of reform in socialist countries. *Journal of Economic Perspectives* 4, Summer, 131–147.
- O. Lange (1938). On the theory of economic socialism. In *On the Economic Theory of Socialism* (ed. by B. Lippincott), Minneapolis: University of Minnesota Press, pp. 55–143.
- I. Macneil (1974). The many futures of contracts. *Southern California Law Review* 47, May, 691–816.
- J. Marschak and R. Radner (1972). *The Theory of Teams*, New Haven, CT: Yale University Press.
- S. Masten and K. Crocker (1985). Efficient adaptation in long-term contracts: take or pay provisions for natural gas. *American Economic Review* 75, December, 1083–93.
- R. McCain (1977). On the optimal financial environment for worker cooperatives. *Zeitschrift für Nationalekonome* 37, 355–84.
- D. M. Nuti (1989). Feasible financial innovation under market socialism. *Journal of Comparative Economics* 1, 85–104.
- J. Persky (1991). Restrospectives: Lange and von Mises, large scale enterprises, and the economic case for socialism. *Journal of Economic Perspectives* 5, Fall, 229–36.
- L. Putterman (1982). Some behavioral perspectives on the dominance of hierarchical over democratic forms of enterprise. *Journal of Economic Behavior and Organization* 3, June, 139–60.
- L. Putterman (1990). *Division of Labor and Welfare*, New York: Oxford University Press.
- J. Robinson (1933). *The Economics of Imperfect Competition*, London: Macmillan.
- S. Rosen (1988). Transaction costs and internal labor markets. *Journal of Law, Economics, and Organization* 4, Spring, 49–64.
- M. Rothschild and J. Stiglitz (1976). Equilibrium in competitive insurance markets. *Quarterly Journal of Economics* 80, November, 629–50.
- D. Sappington (1991). Incentives in principal-agent relationships. *Journal of Economic Perspectives* 5, Spring, 45–66.

- J. A. Schumpeter (1942). *Capitalism, Socialism, and Democracy*, New York: Harper & Row.
- J. A. Schumpeter (1947). The creative response in economic history. *Journal of Economic History* 7, (November), 149–59.
- H. A. Simon (1962). The architecture of complexity. *Proceedings of the American Philosophical Society* 106, December, 467–82.
- H. A. Simon (1985). Human nature in politics: the dialogue of psychology with political science. *American Political Science Review* 79, 293–304.
- B. Ward (1967). *The Socialist Economy: A Study of Organizational Alternatives*, New York: Random House.
- T. Weisskopf (1991). The drive toward capitalism in East Central Europe: is there no other way? Unpublished manuscript.
- O. E. Williamson (1975). *Markets and Hierarchies: Analysis and Antitrust Implications*, New York: Free Press.
- O. E. Williamson (1983). “Credible commitments: using hostages to support exchange”. *American Economic Review* 73, (September), 519–40.
- O. E. Williamson (1985). *The Economic Institutions of Capitalism*, New York: Free Press.
- O. E. Williamson (1988). Corporate finance and corporate governance. *Journal of Finance* 43, July, 567–91.
- O. E. Williamson (1989). Transaction cost economics. In *Handbook of Industrial Organization*, (ed. by R. Schmalensee and R. Wiliy), Amsterdam: North-Holland, pp. 135–82.
- O. E. Williamson (1991a). Comparative economic organization: the analysis of discrete structural alternatives. *Administrative Science Quarterly* 36, June, 269–96.
- O. E. Williamson (1991b). Economic institutions: spontaneous and intentional governance. *Journal of Law, Economics, and Organization* 7 (Special Issue), 159–87.
- O. E. Williamson (1992). Calculativeness, trust, and economic organization. *Journal of Law and Economics*, forthcoming.
- L. Zucker (1986). Production of trust: institutional sources of economic structure, 1840–1920 *Research in Organizational Behavior* 6, 53–111.